

ARCTIC: Empowering Transdisciplinary Engineering Education Through Immersive Augmented Reality Experiences

Kevin KASTNER ^{a,1}, Felix WUEHLER ^a, Florian KOLB ^a, Julian REICHWALD ^a,
Matthias RAEDLE ^a and René EISWIRTH ^b

^a*Virtual Engineering Competence Center, Mannheim University of Applied
Sciences, Mannheim*

^b*Institute for Thermal and Biological Process Engineering, Mannheim University
of Applied Sciences, Mannheim*

Abstract. In the pursuit of modern and future-thinking education, students must be more oriented towards research and the practice-oriented development of products and processes. Experts from various disciplines are currently forced to use complex measurement and equipment infrastructure but are unable to adequately combine and visualize the increasingly complex results from the various fields in a suitable form. To explain these complex phenomena to students as intuitively as possible, the augmented reality app ARCTIC ("Augmented Reality Comes To Interactive Campus") was designed for practice-oriented lectures. This approach enables the projection of (simulation) data into physical spaces using smartphones and facilitates user interaction with 3D models. By using open-source principles, the app promotes collaboration and enables experts from different fields to provide cost-effective solutions for teaching individually. This approach not only improves the accessibility of the app but also promotes a community-oriented, sustainable development model. The app was tested under pedagogical supervision in real conditions with students from various disciplines of chemical and process engineering, as well as AEC companies. The evaluation showed that students and company representatives are extremely open to the developed AR solution. They would strongly support the further integration of the app into teaching. This was also evident from the surveys, in which students participated constructively and contributed further ideas. Augmented reality is a promising technology with the potential to revolutionize teaching. ARCTIC is a successful example of how AR can be integrated into the classroom to explain complex phenomena intuitively and interactively to students.

Keywords. Augmented reality, computational fluid dynamics, educational technology, transdisciplinary. Engineering, open-source

Introduction

Modern education requires a shift towards research and practical application. Although professionals across various disciplines use complex measurement and equipment, effectively combining and visualizing intricate, multi-dimensional data remains a challenge. This paper explores the potential of Augmented Reality (AR) to address gaps

¹ Corresponding Author, Email: k.kastner@hs-mannheim.de.

in educational practice. One of the most widely used tools in engineering is Computational Fluid Dynamics (CFD). It is often used to analyze, design, and optimize chemical engineering processes. CFD is a powerful tool to get insights due to the possibilities for visualization. Strategies for collaborative work represent an underresearched and under-published area [1, 2]. The Corona pandemic shows the importance of such well-designed and prepared workflows for schools and universities, given the rapid shift to online teaching. In addition to its use in teaching, this technology is also becoming increasingly important in industries such as Architecture, Engineering and Construction (AEC) [1]. An elaborated method for integrating CFD simulations to be displayed with AR can make it much easier to get started with complex subjects. The use of AR is justified by the following points: the complementary environment is optimal for teaching; there is the possibility of real-time interaction; complex relationships can be shown in a more comprehensible way; inexpensive hardware; easier accessibility; the capability to load data during runtime allows for immediate processing, resulting in quicker insights. It is important to note that this work combines three disciplines: simulation, preparation, and visualization of 3D data. The tool is open-source and aims to be accessible to all, regardless of their background. Furthermore, the framework, including the innovative converter, prioritizes broad file format compatibility to ensure seamless integration with diverse data sources. This is particularly important because CFD files, due to their fine level of detail (small discretization), often reach significant sizes [3]. But for a fast and lean workflow, the files must be as small as possible to realize the smooth transfer of data and visualization on the selected end device. So, we present ARCTIC ("Augmented Reality Comes To Interactive Campus"), an AR application designed for interactive lectures in practice-oriented fields. ARCTIC utilizes smartphones to project simulation data directly into physical spaces, enabling students to interact with 3D models for a more intuitive understanding of complex phenomena. Building on the previously discussed approaches, this application aims to leverage open-source principles to foster collaboration among experts from diverse fields. This collaborative effort will enable the development of cost-effective and adaptable educational tools. The concept supports the accessibility of the application and a sustainable development model centered on community input. In addition, by addressing the affordability gap in AR technology, this app aims to level the playing field and allow everyone, regardless of income, to experience the benefits of AR. The paper explores how AR has the potential to revolutionize education by using this technology to transform complex subjects into an intuitive and interactive learning experience, making them easier for students to understand. In a case study, ARCTIC is finally presented to demonstrate the power of AR in the classroom under real-world conditions and pedagogical supervision.

1. Related Work

From a cognitive and pedagogical perspective, Augmented Reality exhibits a strong alignment with constructivist learning theories [4]. Solmaz et al. present a method for collaborative work [3, 5]. However, the study also emphasizes a significant challenge: the possibility of integrating such a topic requires a streamlined workflow for instructors. His current application requires expertise, which makes it a hurdle for new users without additional training [6, 7, 8, 3]. The positive effect on the integration of CFD and AR visualizations in the classroom is confirmed in several studies [9, 10, 11, 3, 12, 13, 14,

15, 16]. Students report an improvement in the learning effect when teaching difficult content. This method is well suited to visualize complex and unobservable phenomena, such as fluid flow simulations [17]. Immersive technologies can be used to approximate digital environments. Especially AR, which integrates digital data into the natural environment, offers one possibility [5]. To display AR content various devices can be used for visualization. Differently from Virtual Reality (VR), which requires specialized headsets, AR content can be displayed on easily available devices like handheld devices such as a tablet or smartphones [18, 19, 20]. Also, by utilizing existing devices, eliminating the cost barrier associated with dedicated VR equipment, and potentially impacting a broader audience across various fields. This advantage is because almost every end-user has the necessary hardware at home. Optionally, the presentation can also take place on other end devices, such as Powerwall or even a Cave Automatic Virtual Environment (CAVE) technology, for a more immersive user experience. As a result, the integration of VR and especially AR in didactic areas is becoming increasingly important and has promising potential, especially in the scientific field, where successful experiments have already been conducted [11, 3, 12, 13, 14, 15, 16, 10]. Along with the opportunities of AR, some risks must be taken into consideration. Different users experience different symptoms, which are collectively referred to as Cybersickness (CS). CS is somewhat less common in AR because the real world is still displayed, and the user can orient themselves better than in VR. However, the user may still experience cognitive overload, which can manifest as dizziness and nausea [14, 3, 21, 17, 22]. Research with VR devices aims to capture the characteristics of users in an immersive virtual environment. This often involves the use of biosensors that depend on personal data. Ruichon et al. propose an approach that uses kinematic data and Artificial Intelligence (AI) to predict Cybersickness [23]. Such systems could be advantageous in the future, as they do not require additional equipment and can still ensure a better user experience. Another issue is the influence of the different models on the participants. Chandio et al. showed in their study that presence can be detected by reaction time. They found a significant change in user's reaction time between a realistic and an abstract virtual object [24]. This realization leads to another problem: creating realistic models is extremely time-consuming and costly. It requires deep expertise in areas such as fluid mechanics, materials science, and Computer Aided Design (CAD) modelling. In addition, due to the large number of CFD software programs that use different file formats and structures, these must then be meticulously converted into the appropriate file format to be compatible with AR visualization software [5, 25, 26]. A possible solution to keep the workflow simple could be to minimize the number of software tools used and use open-source programs or frameworks.

2. Materials and Methods

There are many possibilities of Software Development Kits (SDKs) for the development of a mobile app. These interfaces can be used via plugged-and-play in different game engines like Unity or Unreal Engine. Which allows porting AR content on mobile platforms such as Android, iOS or even Head-Mounted Displays (HMDs) and interacting with them in real-time. The app used in the study was made with the cross-compilation platform Unity 2020.3.5f1. To create an AR environment, the AR Foundation framework version 4.0.12 is used, which is officially supported by Unity. The framework itself only offers several interfaces and requires additional packages for the destination target

platform. The app is primarily planned for end devices with Android and iOS Operating Systems (OS). In addition, because Foundation supports the main features of both Google's ARCore as well as ARKit developed by Apple, the application implementation follows a uniform workflow for both platforms. Another advantage of this framework is that it secures the future development and customization of the AR app. The problem with switching between different OSs is that multiple platforms do not always support all functionalities. Foundation therefore keeps a placeholder that can be replaced later if the functionality is supported on the specific platform by an update of the framework without rebuilding the whole project. Finally, Unity has a large (developer) community so there is a lot of information and support for various problems as well as a continuous development of new technologies. Which also fits with the goal of the application to be released open source.

2.1. Converter Overlay

The biggest problem is the integration of AEC data into Unity. The game engine allows portability from the most common 3D programs such as Blender, Maya, or 3ds Max but it does not support native import of Building Information Modeling (BIM) or AEC data like .step, .jt, .wrl, or .vtk. For this reason, a pre-processing of the data by a specially developed data converter is necessary. For this purpose, an intuitive and universal data converter is provided alongside the app. Within the Unity App, the binary representation of the GL Transmission Format .glb is selected as a high-performance and uniform display of 3D models. To integrate the open-source idea into the data converter, it was implemented in Python3 and is equipped with a PySide2 user interface. Pyglet is used for rendering and interacting with the model. This enables the rendering and interaction of OpenGL graphics directly via Python3. The trimesh library is integrated into the backend to process the meshes and convert the data. When selecting the library, particular attention was paid to the possibility of converting .step, .stl and .vtk files to .glb. The converter makes it possible to import one or more AEC files into a scene and then export this scene as .glb. The storage location can be a dedicated FTP server or any local computer with Internet access. Once the conversion is complete, a QR code is generated for the user, revealing the storage location of the data. Once the QR code is recognized by the app, the simulated scenes can be placed in the AR environment. This enables a quick and easy workflow and allows for easy sharing of the generated scenes. An additional challenge is the size of the expected data. Based on this fact the converter not only converts the file but also does an innovative data reduction at the same time to run the simulation data as performant as possible on a wide range of end devices. Optionally, the Python approach allows the converter to be extended with additional features. For example, the integration of openFOAM for the direct CFD calculation within the converter would be conceivable [27].

2.2. Application Overlay

The components of ARCTIC for visualizing Real Time 3D (RT3D) content are designed modular therefore it works like the plug-and-play principle. Which makes future adaptations and an easy integration of new features possible. To place the (simulation) data into the AR environment there are two possibilities. In the first case, the object is placed in the room by touching the display and then interacting with it. The second and more commendable alternative is marker-based applications. AR Foundation offers

different opportunities for layer, face, 2D image, or object recognition as well as tracking in the real world. Figure 1 describes the process of importing simulation data from third-party programs such as ParaView or COMSOL Multiphysics to ARCTIC.

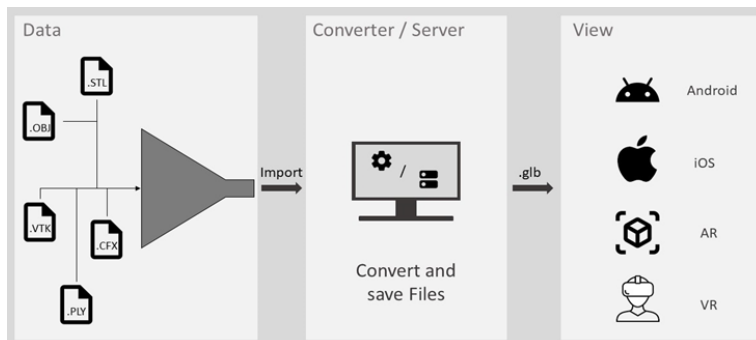


Figure 1. ARCTIC's Workflow: imports various CAD data formats, then optimizes, converts, and stores the data using a proprietary converter in the middle. Finally, the data is visualized on various platforms on the right.

By using the innovative data converter, the app does not require a specific data format instead it supports usual exchange formats which are processed as described above, and the output file is saved centrally for example on a server or local storage. Afterwards, the user creates a QR code that contains all the information to load the simulation data at runtime.

3. Results

3.1. Apparatus

The study was accomplished with bachelor students from the Faculty of Chemical and Process Engineering. Each participant used their own mobile device for the study to communicate via the university's internal network. Additionally, all necessary materials, including tracking QR codes as well as scanning codes, were made available through the teaching platform Moodle. The network provided access to a dedicated server where the lecture content models were stored. Experimenters assisted with application installation and use as needed.

3.2. Task

Instructors integrated the pre-created codes into their presentation slides. Whenever a model was required for an explanation, participants could scan it using the app and place it either in front of them or on the presentation surface with the tracking code in the room. After being introduced to the model, they were given sufficient time to interact with it independently. In case it was a chemical and process lecture various flow patterns and heat transfer phenomena of a shell-and-tube heat exchanger can be observed, as shown

in Figure 2. Participants can discuss these observations with each other and ask topic-specific questions that can be explained based on the visualization.

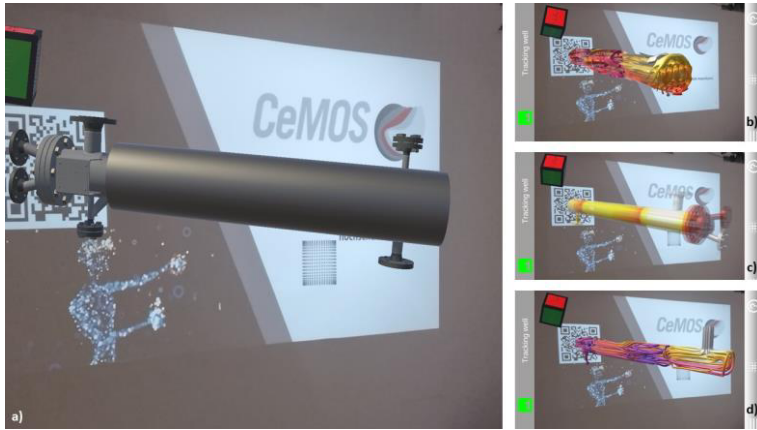


Figure 2. Example of a model used during the experiment. a) On the left is the in-housed shell-and-tube heat exchanger and b)-d) some insights with different simulations with b) isosurfaces c) temperature flow of the shell d) streamlines.

After the lesson was completed, all participants were requested to take part in a post-test evaluation through an online survey. The survey aimed to assess the participant's performance using a scholastic grading paradigm that ranged from one to six. A score of one indicated the highest ranking, while six indicated the lowest. Interrogative themes spanned the app's interface structure, presentation quality, and the pedagogical efficacy of augmented reality within the university context.

3.3. Subjects

A total of 41 subjects participated in the experiment. All participants were in the 3rd or 4th semester of the bachelor's course heat and mass transfer (transport processes) or thermal process engineering, which is part of the faculty of chemical and process engineering at the University of Applied Sciences Mannheim. The age of most participants is between 20 and 25 years, although the age range extends from 18 to over 30 years. The participant's gender was not recorded.

3.4. Result

The results show an overall positive perception of the application by the users, with the average grade for all criteria lying between 1.27 and 2.66.

Asking the participants about how good the graphical user interface (GUI) fitted on their phone and asking if they find the technology forward looking for educational applications received the best average score. Here, future orienting received an average of 1.27 while the GUIs fit was also rated favorably with an average of 1.46 and 92.68% of respondents rating it at least grade 2. The relative amount of students rating each criteria at least grade 2 is displayed in Figure 3. The augmented and virtual reality (AR/VR) aspects were rated constructively. Both questions like if they find AR/VR interesting or if they could imagine using more AR/VR received a positive rating with an average of 1.46 and 1.54 respectively and approval rates of 97.56% and 87.80%.

Although some criteria achieved lower average scores, these were also predominantly rated positively. The scanning quality criterion received an average score of 2.24, with 85.37% of respondents giving it a score of at least 3. Users rated the application positively overall. In particular, the user interface, AR/VR functionalities and future-oriented features were rated positively.

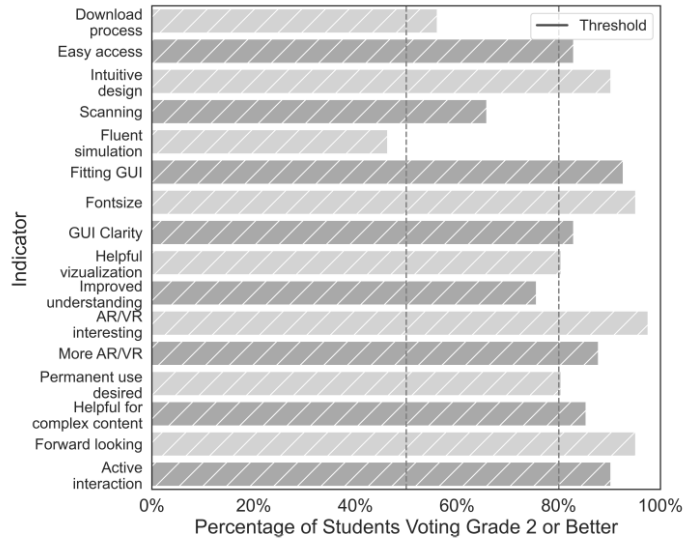


Figure 3. Evaluation results showing relative amount of students rating a criteria at least grade 2. Better

4. Discussion

The insights garnered from our survey during the alpha phase of application development offer a comprehensive understanding of user perceptions and experiences. Through rigorous evaluation, encompassing responses from professors, students, and lecturers at Mannheim University of Applied Sciences, we have gained valuable quantitative insights into various dimensions of the application's performance.

The data reveal a generally positive reception towards the application, with participants expressing favorable sentiments towards key features. Notably, dimensions such as the completeness of the GUI, adequacy of font size, and interest in AR/VR models received particularly high ratings, with mean scores below 1.5. These findings underscore the application's potential as an effective educational tool, with users demonstrating enthusiasm for features that enhance user interface clarity and facilitate immersive learning experiences.

However, a closer examination of the data highlights areas where enhancements may be warranted. For instance, aspects such as the download process and marker scanning received slightly lower ratings, with mean scores closer to 2.5. While still indicative of overall satisfaction, these findings suggest opportunities for optimization to streamline user interactions and enhance usability. Addressing these concerns is pivotal to ensuring a seamless user experience and maximizing the educational impact of the application.

In response to these findings, we propose targeted strategies for improvement. Efforts to refine the download process and optimize marker scanning capabilities should

be prioritized, drawing upon user feedback to inform iterative enhancements. Additionally, further exploration of ways to enhance simulation fluidity and integration of AR/VR models into lectures are recommended avenues to enrich the learning experience.

By embracing a user-centric approach and leveraging quantitative insights from our survey data, we are committed to iteratively refining the application to meet the evolving needs of our users. Through ongoing optimization and innovation, we aim to develop a robust and user-friendly educational tool that empowers learners in the field of process engineering to engage with complex concepts effectively and efficiently.

In order to enhance the appeal and currency of engineering courses, it is necessary to incorporate modern (AR) technologies. These technologies encourage students to engage actively in the lecture, enabling them to design the course and its focus individually. This approach facilitates individualized and self-directed learning, regardless of the student's level of knowledge. For instance, as illustrated here, students can view the 3D model of a heat exchanger from all angles, thereby enhancing their comprehension of the heat exchanger's functionality and structure. These advantages of AR can be transferred not only to the domain of education but also to the future working environment or industrial context. The future specialists can utilize AR applications on tablets or smartphones to access relevant information in real-time, thereby enabling them to work more expeditiously and efficiently. This can be employed, for instance, in the maintenance of machines or the troubleshooting of complex systems.

However, the technology is not only useful in engineering itself, but also in transdisciplinary dialogue between interface positions. A modern visualization of facts promotes the understanding of all those involved, especially those who have less technical understanding or spatial imagination.

4.1. Limitations

Firstly, it is important to highlight that the current version of the application lacks support for iPhones, limiting accessibility for users who predominantly use iOS devices. The exclusion of iPhone users not only diminishes the reach of the application but also hampers efforts to foster inclusivity and equity in educational technology. The use of 3D models presents a general issue. Since the models are university-specific, there are no universally valid models or open-source alternatives available. Instead, the specific components must be reconstructed using CAD, which is a time-consuming and technically demanding process that requires special expertise. Furthermore, the application only uses .glb files due to their compatibility with web-based AR platforms. Accordingly, the data to be displayed is converted from the original format to .glb during the usage process. Conversely, however, this means that in most cases the application is dependent on an upstream work step, as the industry-standard file formats differ. Lastly, it's important to recognize that the functionality of the application may be constrained by hardware limitations, particularly in the case of Google AR support. Not all devices are equipped with the necessary sensors and processing power to fully leverage Google's AR capabilities, which may result in suboptimal user experiences for certain users.

4.2. Lessons learned

Considering the aforementioned limitations, future research endeavours should aim to address these constraints by exploring other importing options, enhancing cross-platform

compatibility, and optimizing the application's performance across a wider range of devices. By mitigating these limitations, we can foster broader adoption and utilization of AR technologies in educational settings, ultimately enhancing learning experiences for a more diverse range of learners. However, the survey results also show that the application helps students to learn spatial visualization. For some people, working with three-dimensional objects can cause difficulties that can be overcome by using the application. Nevertheless, social imbalances must be considered when using the application in teaching. For example, differences in wealth, which are also reflected in the hardware used, can play a major role.

5. Conclusion

Augmented reality technologies have the potential to revolutionize the way we teach by making it possible to practically visualize abstract concepts and complex relationships through AR apps like ARTIC. This helps bridge the gap between theory and practice and promotes transdisciplinary collaboration. AR is also democratizing access to education and expertise, leading to new forms of collaboration and knowledge sharing. Learners can not only passively receive complex topics but also actively explore, interact with, and understand them. Thanks to the open interfaces, it is possible to visualize a wide range of subjects, from process simulations in science, technology, engineering and mathematics (STEM) to the reconstruction of historical buildings in the humanities. Additionally, the simple integration into existing teaching environments, according to the motto "Bring your own device", minimizes administrative efforts for teachers and IT staff. However, the study also revealed challenges associated with AR in teaching. These challenges include digital imbalance, technical obstacles related to smartphone compatibility, and integrating such media into didactics. Even with these challenges, both learners and industry provided positive feedback. Therefore, the development of AR content can be seen as a forward-looking step that has the potential to revolutionize not only the education sector but also improve access to knowledge for all.

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